24K0756 DAA Assignment #2. BCS-5A Date:

Q1 :

1. Take every edge of polygon A & compare it with every edge of polygon B.
2. check whether the two line segments intersect.
3. If any 2 edges intersect, return true.
4. If no edges, intersect, check whether one vertex of A is inside B.
5. If not, check whether one vertex of B is inside A.
6. If either vertex is inside the other polygon, return true.
7. Otherwise return false.

Justification :- Checking only edge crossings is insufficient b/c one polygon maybe completely inside another w/o crossing its boundary.

Q2 :

1. Take all the vertices of star shaped polygon.
2. Find the point with lowest y-coordinate & make it ref. point.
3. Sort remaining points by polar angle around a reference (ref.) point.
4. Apply Graham-scan method, by considering points in turn & skipping the middle point if turn isn't ccw.
5. Return the remaining vertices in coordinates order.

Justification:- The kernel helps determine visibility within a star shaped polygon, but it does not change the definition of convex hull. The hull must enclose all ~~values~~ vertices, including those

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in inward facing parts of polygon. Standard convex hull algo. don't need a special mod. merely b/c the polygon is star shaped. The input points can be processed normally with care of for duplicate & colinear points.

Q3: Method 1:-

1. Draw a horizontal ray from query point towards the right
2. Count the no. of times the ray ~~becomes~~ crosses polygon edges.
3. If the no. of crossings is odd, the point is inside & vice versa.

~~4~~ - Before Count

- Before counting if point lies of edge exactly. If it does, classify it as boundary.

Method 2:-

1. Draw aline from the query point to each polygon vertex.
2. Calculate the signed angle b/w consecutive lines.
3. Add all signed angles.
4. If total is approx. $\pm 360°$, point is inside.
5. If the total is approx. $0°$, the point is outside.

- Boundary points must be checked seperately.

Q4:

Suppose a set of points is spread aeross a region & forms several seperate groups.

1. Read all points.
2. Apply a clustering algo. such as DBSCAN or distance-based clustering, to divide the points into groups.

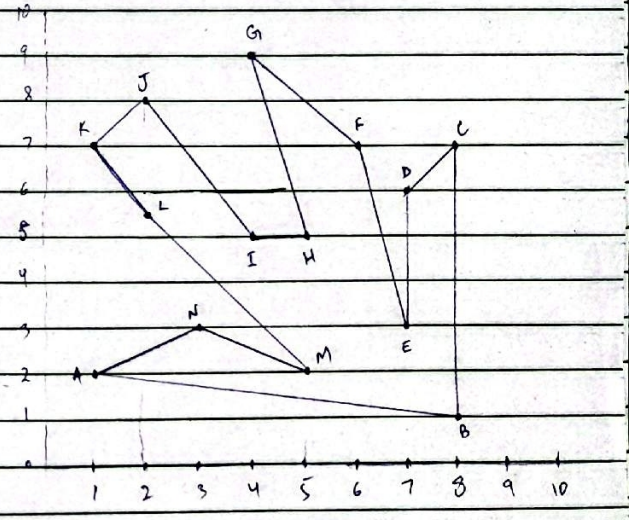
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3. For each group calculate its convex hull using grahams scan.
4. Compare the convex hull of different groups.
5. If the hulls overlap or touch, determine whether the groups are actually disjoint or whether the clustering threshold has caused a misleading seperation.
6. If the hulls are seperate & ~~the~~ each group forms a convex region, report the ~~seperate~~ convex polygons.

- Clustering identify groups of neighboring points. Convex hull computation then gives the smallest convex boundary around each group.

Q5:

A(1,2) B(3,1) C(8,7)
D(7,6) E(7,3) F(6,7)
G(4,9) H(5,5) I(4,5)
J(2,8) K(1,7) L(2,5.5)
M(5,2) N(3,3)



CamScanner

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(i) Brute Force Approach

starting Point	Pair to Test.
A	AB, AC, AD, AE, AF, AG, AH, AI, AJ, AK, AL, AM, AN
B	BC, BD, BE, BF, BG, BH, BI, BJ, BK, BL, BM, BN
C	CD, CE, CF, CG, CH, CI, CJ, CK, CL, CM, CN
D	DE, DF, DG, DH, DI, DJ, DK, DL, DM, DN
E	EF, EG, EH, EI, EJ, EK, EL, EM, EN
F	FG, FH, FI, FJ, FK, FL, FM, FN
G	GH, GI, GJ, GK, GL, GM, GN
H	HI, HJ, HK, HL, HM, HN
I	IJ, IK, IL, IM, IN
J	JK, JL, JM, JN
K	KL, KM, KN
L	LM, LN
M	MN

We pick 6 pairs AB, BC, CG, GJ, JK, KA

For pair AB:

$S = (x_2 - x_1)(y_3 - y_1) - (y_2 - y_1)(x_3 - x_1)$

$S = (8-1)(y-2) - (1-2)(x-1) \Rightarrow 7(y-2) + (x-1)$

C(8,7) = 42, D(7,6) = 34, E(7,3) = 13, F = 40, G = 52, H = 25

I = 24, J = 43, K = 35, L = 25.5, M = 4, N = 9

Since all dist.(s) are one positive so 'AB' is valid.

For pair BC:

$S = (8-8)(y-1) - (7-1)(x-8) \Rightarrow -6(x-8)$

A = 42, D = 6, E = 6, F = 12, G = 24, H = 18, I = 24, J = 36, L = 36, M = 18, N = 80

All +ve, BC included.

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- Pair CG: A = 34, B = 24, D = 6, E = 18, F = 4, H = 14, I = 16, J = 8, K = 14, L = 18, M = 26, N = 26. All +ve, CG included.
- Pair GJ: A = 11, B = 20, C = 8, D = 9, E = 15, F = 6, H = 9, I = 8, K = 1, L = 5, M = 15, N = 11, All +ve, GJ included.
- Pair JK: A = 5, B = 13, C = 7, D = 7, E = 10, F = 5, G = 1, H = 6, I = 5, L = 2.5, M = 8, N = 6, All +ve, JK included.
- Pair KA: B = 35, C = 35, D = 30, E = 30, F = 25, G = 15, H = 20, I = 15, J = 5, L = 5, M = 20, N = 10.

A → B → C → G → J → K → A

$O(n^2)$ → for pair & $O(n)$ for other points.

Time complexity = $O(n^3)$

(ii) Quick Hull Approach.

left-most → A = 1,2 , Rightmost → B = (8,1)

cross = (A, B, D) = $x + 7y = 15$

- Find farthest point from AB.

C = 42, D = 34, E = 13, F = 40, G = 52, H = 25, I = 24, J = 43, K = 35, L = 25.5, M = 4, N = 3

Farthest point = G = 52 → G(4,9).

Points on outside of triangle ABG on the AG side.

Points on outside of triangle ABG on the GB side.

Cross(A, G, J) = 18

Cross(A, G, K) = 15 ⇒ K selected.

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Cross (K, G, J) = (4-1)(8-7) => -(9)(7)(2-1) => 1

since no remaining outside points.

Process for GB

Cross (GBC) = (8-4)(7-9) => -(1-9)(8-4).

= 24

C is selected

G → C → B

- A → B → C → G → J → K → A
- Time Complexity = O(nlogn).

Graham's Scan;

(1) Selected the point with min. y-coordinate. B(8,1).

(2) Calculate Polar angles & sort points.

∠BC = 90° , ∠BD = 101.51° ∠BF = 108.43° ∠BE = 116.57, ∠BH = 126.87°
∠BJ = 130.60° ∠BI = 135° ∠BK = 139.4° ∠BL = 143.13° ∠BN = 158.20°,
∠BM = 161.57° ∠BA = 171.87°

B → C → D → F → E → H → J → I → K → L → N → M → A

- Stack = [B, C]

Cross.(B, C, D) = 0(5) - 6(-1) = 6

- Stack = [B, C, D]

Cross (C, D, E) = -1(1) - (-1)(-1) => -2 Remove 'D'

Cross (B, C, P) = 0(6) - (6)(-2) => -12

- stack = [B, C, F]
  Cross (C, F, E) = -2(-4) - 0(1) = 8

- Stack = [B, C, F, E]
  Cross (F, E, G) = 1(6) - (-4)(-3) => -6 (Drop E)
  Cross (C, F, G) = (-2)(2) - 0(-2) => -4 (Drop F)
  Cross (B, C, G) = 0(8) - 6(-4) => -24.

- Stack = [B, C, G]
  Cross => (C, G, H) = -4(-4) - 2(1) = 14

- Stack [B, C, G, H]
  Cross (G, H, J) = 1(3) - (-4)(-3) = -9 (Drop H)
  Cross (C, G, J) = (-4)(-1) - (2)(-2) = 8

- Stack [B, C, G, J]
  Cross (G, J, B) = -2(-3) - (-1)(2) = 8

- Stack [B, C, G, J, I]
  Cross (J, I, K) = 2(2) - (-3)(-3) = -5 Drop I
  Cross (G, J, K) = -2(-1) - (-1)(-1) = 1

- Stack [B, C, G, J, K]
  cross (J, K, L).
  :
- Stack [B, C, G, J, K, A]

Question 6:

A(-1,0) B(1,3), C(3,5), D(5,4), E(7,1), F(5,-3), G(2,-4)
H(-2,-2), I(2,1), J(3,0), K(1,7) L(4,2).

left most: H(-2,2)
Right most: E(7,1).

Cross (H,E,P):- $9(y+2) - 3(x+2)$.
$= 3y - x + 4$

A=5, B=12, C=16, D=11, F=-10, E=-10, G=-10,
I=5, J=1, K=6, L=6

- Positive = A, B, C, D, I, J, K, L
- Negative = G, F
- Farthest point from Possible side is 'C'.
- Recursively partition for +ve side.
- Cross (H,C,P) = 5y - 7x - 4

A=3, B=4, C=-9, I=-3, J=-15
K=-6, L=-12.

Since B are +ve, outside from HC & farthest is B.

- Cross (C,E,D) = 4(x+y-8)

A=-36, B=-16, D=-44, I=-8, J=-20, J=-20, K=-24
Since D is +ve outside the hull.

- From triangle EFH
Cross (E,F,P) = 4x - 2y - 26
G=-10, H=-30; Since all are -ve those mean are
- Cross (F,H,P) = -x - 7y - 16
E=-30, G=10.
- Cross (F,G,H) = -10
- E → F → G → H
H → A → B → C → D → E → F → G → H.

Question 7:

(i) Pairwise Analysis :-

S1 = (1,1) → (5,5) on line y=x
S2 = (1,5) → (5,1) on the line y = 6-x
S3 = (5,5) → (5,1) on line x=5
S4 = (2,2) → (4,4) on line y=x
S5 = (0,0) → (0,4) on line x=0.

Unique Pair E_2^5 = 20 pairs.

S1-S2	y=x, y=6-x meet at (3,3) inside both	Proper Crossing.
S2-S4	same point (3,3) inside S2 (x,15-x) & S2(x,2-4)	" "
S1-S3	S1 reaches x=5 at (5,5), the endpoint of both	Endpoint Intersection
S2-S3	S2 ends at (5,1) also an endpoint of S3	" "

S1-S4	Colinear (both on $y=x$), S4 lies entirely within S1	Colinear overlap along (2,2)-(4,4)
S1-S5	line $y=x$ meets $x=0$ at (0,0), but S1 starts at $x=1$	No intersection
S2-S5	S2's x-range is 1-5, so it never reaches $x=0$	No intersection
S3-S4	S4's x range is 2-4, so it never reaches $x=5$	" "
S4-S5	S5's x range is 2-4, so it never reaches $x=0$	" "

Proper ~~intersection~~ crossing = (S1,S2), (S2,S4), both at (3,3).
Endpoint Intersection = (S1,S3) at (5,5) and (S2,S3) at (5,1)
Colinear Overlap = (S1,S4)
Non-Intersecting = (S1,S5), (S2,S5), (S3,S4), (S3,S5), (S4,S5).

(ii) Brute Force Algorithm :-

Orientation test : $orient(p,q,r) = (q.x-p.x)(r.y-p.y) - (r.y-p.y)(r.x-p.x)$

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intersecting Pairs (S[1..n]) {
for(int i=1 to n-1):
    for j=i+1 to n:
    (P1,Q1) = S[i] ; (P2,Q2) = S[j]
    O1 = orient(P1,Q1,P2)
    O2 = Orient(P2,Q2,Q1)
    O3 = Orient(P2,Q2,P1)
    O4 = Orient(P2,Q2,P1)
if.(O1*O2 && O3*O4 < 0):
    return (i,j) "Proper Crossing"
else if O==0
    return "colinear"
return "endpoint touch".
```

Each pair needs O(1) work
Number of pairs = $n(n-1)/2$
$T(n) = c \cdot n(n-1)/2$
Time Complexity = $O(n^2)$.

(iii) Scaling to 100000 Locations

Limitations of brute force:

- $n = 10^5$ gives about 5×10^9 pair tests per snapshot.
- At about $10^8 - 10^9$ tests per second, this takes roughly 5-50 seconds per frame which is too slow per real time accident detection.
- Quadratic growth means 10x more data costs 100x more time.
- Most pairs far a part & can never intersect, so the work is wasted.

Bentley-Ottman Sweep Line:

- A vertical line sweeps left to right. Events are segments endpoints & intersecting point, held in priority queue.
- The sweep status is balanced BST that orders the segments currently crossing the sweep line.
- Only segments adjacent in the BST are tested, i/e 2 segments must become neighbors before they cross.
- Time = $O((n+k)\log n)$; k = no. of crossing, space = $O(n+k)$.

Question 8:

$n = 100,000 (10^5)$

$h = 100$

Brute force = $O(n^3)$, calculation = $(10^5)^3$, operation = 10^{15}

Graham scan = $n \log_2 n$, $10^5 \times 16.61$, operation = 1.66×10^6

Jarvis March = $n \cdot h$, $10^5 \times 100$, operation = 10^7

- Graham scan vs b.f = $10^{15} / 1.66 \times 10^6 = 6 \times 10^8$ times fewer operations
- Jarvis vs B.f => $10^{15} / 10^7 = 10^8$ times fewer.
- G.S vs J.M => $10^7 / 1.66 \times 10^6 = 6$ times → Graham is faster.

* Choose Graham scan method, its O(nlogn) bound holds regardless of the hull size, so it is predictable & fast.
- Jarvis is also acceptable if it known to be tiny ($h < \log n$)

Implementation errors :-

- The core test is the cross product orientation :-
  $(q.x - p.x)(r.y - p.y) - (q.y - p.y)(r.x - p.x)$
- with 32 bit int - coord(s) products can exceed 2^{31}, causing wraparound → wrong sign & wrong turn direction, this can produce non convex or broken hull.
- with f.p coord.(s), rounding makes nearly collinear pts appear as left or right turns causing inconsistent decisions.
- Fix use 64 int.

Duplicate integers points :

- Identical points give a cross product of 0 & non-zero length vectors, Graham scan's polar angle sort becomes ambiguous.
- Jarvis may loop forever or select the same pt. repeatedly when it picks the most "counter clockwise" candidate.
- Remove duplicates first (sort + unique) so that every point is distinct

Collinear pts when three pts have cross product exactly 0, the algo. must decide whether to keep the middle point.

Question 9:-

Ray Casting Intersection :

A(1,1), B(7,1), C(9,4), D(7,8), E(4,7), F(2,9), G(0,5).

Suspicious = P(4,4), Q(8,6), R(10,5), S(4,1) $(x_1, y_1)(x_2, y_2)|(x_p, y_p)$.

$$x_{int} = \frac{x_1 + (y_p - y_1)(x_2 - x_1)}{y_2 - y_1}$$

(pt)

if $x_{int} < x_p$, intersection lies to the right of the test point & it is counted.

- Pt.(4,4)

  $y = 4$, $y \le 4 <$ max.

			(Range of 4)
AB	(1,1) → (7,1)	y=1	No, Parallel to ray.
BC	(7,1) → (9,4)	1-4	4 < 4 false, No
CD	(9,4) → (7,8)	4-8	4 ≤ 4 < 8, yes

DE	(7,8) → (4,7)	7-8	No
EF	(4,7) → (2,9)	7-9	No
FG	(2,9) → (0,5)	5-9	No
GA	(0,5) → (1,1)	1-5	1≤4≤5, yes.

CD & GA are selected, now check whether they are right of P.

CD (9,4) D(7,8)

$x = 9 + (4-4)(7-8)/(8-4)$

$= 9 + 0$

$x = 9$

$9 > 4$ select (CD)

G(0,5), A(1,1).

$x = 0 + (4-5)(1-0)/1-5$

$= 0 + 0.25$

$= 0.25$

$0.25 < 4$ reject (GA)

Point Q(8,6)

Consider edge CD with C(9,4), D(7,8).

$y = 6$

$x_{int} = 9 + \frac{(6-4)(7-9)}{8-4} = 9 - 1 = 8$

intersection is (8,6) exactly equal to the test point Q.

Q(8,6) is on the boundary.

R(10,5)

Draw horizontally ray from R(10,5) towards right

$y = 5$

edge CD : C(9,4) → D(7,8)

$x_{int} = 9 + (5-4)(7-9)/8-4 = 8.5$

$8.5 < 10$ (No intersection)

edge GA : G(0,5) → A(1,1).

$x_{int} = 0 + (5-5)(1-0)/1-5$

$= 0$

$0 < 10$ (No intersection)

R(10,5) is outside the polygon. (crossing = 0)

• S(4,1)

$y = 1$

check AB : A(1,1) → B(7,1)

$y = 1$

cross (A,B,S) $= (7-1)(1-1)8 - (1-1)(4-1)$

$= 6(0) - 0(3)$

S(4,1) is on boundary

P(4,4) : Inside.

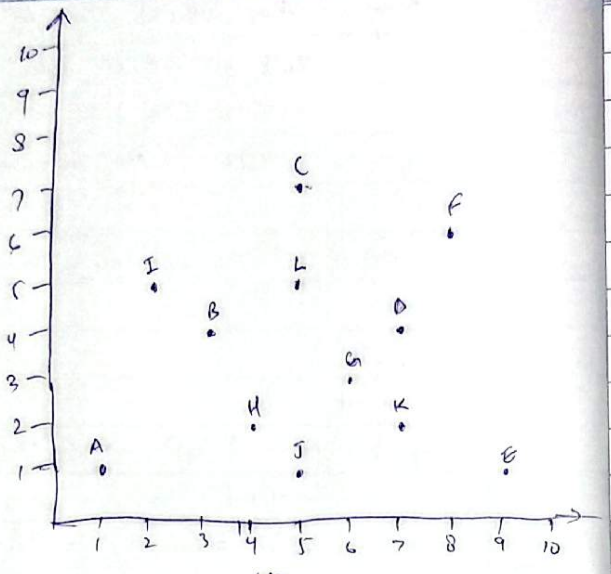
Q(8,6) : Boundary

R(10,5) : Outside

S(4,1) : Boundary.

Question 10:

A(1,1) , B(3,4)
C(5,7) D(7,4)
E(9,1) F(8,6)
G(6,3) H(4,2)
I(2,5) J(5,1)
K(9,2) L(5,5)



(1)

(2) Leftmost : A(1,1)
Rightmost : E(9,1)
Lowest : A(1,1), E(9,1), & J(5,1)
Highest : C(5,7)

(3)&(4) Brute Force :-

Orient(P,Q,R) = $(x_2 - x_1)(y_3 - y_1) - (y_2 - y_1)(x_3 - x_1)$

- Edge AE :

Orient(A,E,P) : $8(y-1)$

Since all points have $y \geq 1$, so AE is hull edge.

- Edge EF :

Orient(E,F,P) : $-8(x-9) - (y-1)$

Since all x values are ≤ 9 so it satisfies orient ≥ 0, EF is hull edge.

Edge FC :

Orient FC(F,C,P) = $-3(y-6) - (x-8)$

all points satisfies that orient > 0; FC is hull edge

Edge CI :

Orient CI(C,I,P) : $-3(y-7) - (2)(x-5)$

all points satisfies orient > 0; CI is hull edge

Edge IA :

Orient IA(I,A,P) = $-(x-2) + 4(y-5)$

All points satisfies orient > 0; IA is hull edge.

Convex hull :- A → E → F → C → I → A

(5) Graham Scan :-

Smallest point :- A(1,1).

Point	J	E	K	H	G	D	F	L	B	C	I
slope	0	0	0.17	0.33	0.4	0.5	0.71	1	1.5	1.5	4

Order : A, J, E, K, H, G, D, F, L, B, C, I

Stack : [A, J, E]

Cross (J, E, K) = 4 > 0 , Push K.

Stack : [A, J, E, K]

Cross (E, K, H) : 3 > 0 , Push H

Stack [A, J, E, K, H] Drop
cross (K, H, G) = -3 < 0 (Pop H).
cross (E, K, G) = -1 < 0 (Drop K)
cross (J, E, G) = 6 > 0 (Push G)

Stack [A, J, E, G]
cross (E, G, D) = -5 < 0 (Drop G).
cross (J, E, D) = > 0 (Push D)

Stack [A, J, E, D]
cross (E, D, F) = -7 < 0 Drop D
cross (J, E, F) > 0 Push F.

Stack [A, J, E, F]
cross (E, F, L) = 16 > 0 Push L.

Stack [A, J, E, F, L]
cross (F, L, B) = 1 > 0 Push B.

Stack [A, J, E, F, L, B]
cross (L, B, C) = -4 < 0 ~~Drop~~(Pop B)
cross (F, L, C) = -6 < 0 (Pop C).
cross (E, F, C) = > 0 (Push C)

Stack [A, J, E, F, C]
cross (F, C, I) = 9 > 0 (Push I)

⇒ Stack [A, J, E, F, C, I]

. .

(6) Comparison :-
B·F = A, E, F, C, I (J on edge AE)
G·S = A, E, F, C, I (J removed by collinear rule)
Q·H ~~J·M~~ = A, E, F, C, I

(7) Time Complexity :-
B·F = $O(n^3)$
G·S = $O(n \log n)$
Q·H = $O(n \log n)$ avg. ; $O(n^2)$ worst case

(8) Shoelace Area :-

	$x_i y_i - y_i x_{i+1}$
A → E	(1)(1) - (1)(9) = -8
E → F	9(6) - 1(8) = 46
F → C	8(7) - 6(5) = 26
C → I	5(5) - 9(2) = 11
I → A	2(1) - 5(1) = -3

Sum = 72
A = ½ |72|
A = ~~72~~ 36 square units.